EX No: 4
13-2-26

Probabilistic Classification using Naive Bayes (Gaussian & Multinomial Naive Bayes)

Pre lab:-

1) Why is Naive Bayes called a probabilistic classifier?

Naive Bayes is called probabilistic classifier because it predicts the class based on probability values using Bayes Theorem. It calculates the probability of each class given the input features and selects the class with the highest probability.

2) What is "naive" assumption made by Naive Bayes?

The naive assumption is that all features are conditionally independent of each other given the class label. In simpler words, one feature does not influence another.

3) How does conditional independence simplify computations?

Conditional independence allows us to compute

$P(x_1, x_2 \ldots x_n | c) = P(x_1|c) \times P(x_2|c) \times \ldots \times P(x_n|c)$

instead of calculating complex joint probabilities we multiply simple individual probabilities. This reduces computational complexity significantly.

4) What is the role of prior probability in classification?

Prior probability $P(c)$ represents the initial likelihood of a class before seeing any data. It helps adjust predictions based on how frequent a class is in the dataset.

5) Why is laplace smoothing necessary?

Laplace smoothing prevents zero probability problems. if a feature never appears in trainning data for a class, its probability becomes zero, which makes the whole result zero. smoothing adds a small value (usually 1) to avoid this issue

6) How does Naive Bayes handle high dimensional data efficiency?

Because of the independence assumption. Naive Bayes computes probabilities seperately for each feature. it does not require complex classification. So it works efficiently even with thousand of features

7) Difference b/w gaussian, multinomial and Bernoulli Naive Bayes?

Types	used for	data type
gaussian NB	continuous data	Assume normal distribut
Multinomial NB	Text classification	word counts
Bernoulli NB	Binary features	0 or 1 (word present/absent)

8 why is Naive Bayes often effective for text classification?

* Text data high dimensions (many words)
* words are approximately independent
* Fast trainning and prediction
* works well even with small dataset.

Post lab
scenario-1
Multinomial Naive Bayes (SMS spam detection)

1) which words contributed most to spam detection?
words like "free", "win", "offer", "urgent", "claim" "prize", "call", "crash" contributed most. These words had high conditional probability in the spam class compared to Ham, so they strongly influenced spam prediction.

2) How did vectorization influence model performance?
Vectorization (count Vectorizer / TF-IDF) converted text into numerical features
* count Vectorization works well for frequency based spam detection.
* TF-IDF reduce the impact of very common words and improves generalization. Better feature representation improved precision and F1-score.

3) what effect did laplace smoothing have?
Laplace smoothing prevented zero probability problems for unseen words. It improves model stability and slightly increased recall by allowing rare words to contribute to prediction instead of eliminating probability

4) Naive Bayes effective for text classification?
* Text data is high-dimensional
* words are aproximately independent
* Trainning is fast and requires less data
* handles sparse matrix efficiently

Hence, it peforms very well for spam filtering.

Scenario-2 - gaussian Naive Bayes
(Iris classification)

1) Why is gaussian NB suitable for numeric features?

Gaussian NB assumes features follow a normal (gaussian) distribution. Iris features (sepal length, petal width, etc.,) are continuous numeric values, so gaussian NB models them using mean and variance.

2) How did feature scaling affect predictions?

Feature Scaling (standard scalar) made all features have equal contribution. It slightly improved accuracy and stabilized probability estimates, especially when features had different ranges.

3) What insight were obtained from class probabilities?

Class probabilities showed the confidence level of predictions. For example, setosa often had very high probability showing it is easily seperable, while versicolor and virginica had closer probabilities due to overlapping features

4) Compare gaussian NB with another classifier (logistic regression)

+ gaussian NB: Faster, probabilistic, works well with small data

+ logistic regression: No independence assumption handles correlated features better.

Conceptual Question:-

1) Why is Naive Bayes computationally efficient?

Because it assumes feature independent it calculates probabilities seperately and multiply them, avoiding complex joint probability computation.

2) What are the limitations of the independence assumption?

In real data, features are often correlated This assumption may reduce accuracy if strong feature dependencies exist.

3) How does Naive Bayes perform with correlated features?

Performance may slightly decrease because correlated features double-count evidence. however, it still performs surprisingly well in many real world tasks.

4) Why is the confusion matrix important in classification?

It shows True positive, False positive False Negative, True Negative helps analyze model errors and compute precision Recall and F1 score

5) How can probabilistic classifier support decision making in real world systems?

They provide confidence scores

(probabilities) not just labels. This helps systems like spam filters, medical diagnosis and fraud detection make risk-based decisions

Description	~~Max~~ mark	Obtained mark
Pre lab	5	3
In lab	10	10
Post lab	5	3
Viva	10	6
Total	30	22
Faculty signature		